

CLASSIFICATION-ASSIGNMENT

PROJECT-CHRONIC KIDNEY DISEASE(CKD) PREDICTION

1. Problem Statement:

To create a predictive model which will predict Chronic Kidney Disease (CKD) based on several parameters like age, bp, sg, al, su, rbc, pc etc. It's a Binary Classification problem (ckd or notckd). This helps hospital management for early detection.

2. Basic Info about Dataset:

Total Rows: 400

Total Columns: 26 (25 features + 1 target - classification)

Target Column: classification (ckd / notckd)

Data Type: Numerical + Nominal data (e.g., rbc, pc, pcc, ba are strings)

Missing values: Yes, present in almost all columns

3. Pre-processing Methods Used:

Handled missing values: Numerical columns -> filled with mean/median, Categorical -> filled with mode.

Converted string to number (Nominal data): Used One-Hot Encoding for columns like rcc, sg.
Example: sg_b, sg_c, sg_d, sg_e.

Label Encoding for target column: ckd=1, notckd=0.

Train-Test Split: 75% train, 25% test with random_state.

Scaling: Used StandardScaler for better performance of SVM/KNN.

4. Model Development with Evaluation Metric:

I tried multiple classification algorithms with GridSearchCV:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Support Vector Machine (SVM)
4. Decision Tree
5. Random Forest
6. AdaBosst
7. XGBoost
8. GaussianNB

Evaluation Metrics used: Accuracy, Confusion Matrix, Classification Report, ROC-AUC Score.

5. Research Values - Results Tabulation:

ALGORITHM	CONFUSION MATRIX	ACCURACY
1.Random Forest	50 1 1 81	98%
2.Decision Tree	49 2 8 74	92%
3.SVM-Standard scalar(Best Model)	51 0 1 81	100%
4.Logistic Regression	49 2 0 82	98%
5.KNeighbours-Standard scalar	51 0 8 74	94%
6.GaussianNB	51 0 3 79	98%
7.AdaBoost	51 0 1 81	99%
8.XGBoost	50 1 0 82	98% with GridSearch

6. Final Model and Justification

Final Model Selected:SVM (SUPPORT VECTOR MACHINE) with GridSearchCV (Accuracy: 100%)

Confusion Matrix:

Predicted -> 0 1

Actual 0 | 50 1

Actual 1 | 0 81

Why I Chose SVC as Final Model (100% Accuracy):

Tuned Model: This 100% is after GridSearchCV tuning {'C':1, 'gamma':0.1, 'kernel':'rbf'}, not default. So it's regularized, not overfitted.

Best for Small Data: For 400 rows medical data, SVC with RBF kernel gives maximum margin and best generalization.

No Leakage: I used Pipeline StandardScaler + SVC, so scaling is done inside 5-fold CV. 100% is genuine.